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ORIGINAL RESEARCH ARTICLES

Integrative Analysis of miRNA and CpG Methylation with the Transcriptome of Lung-Derived Fibroblasts Identifies Epigenetically Regulated Genes in Idiopathic Pulmonary Fibrosis

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Idiopathic pulmonary fibrosis (IPF) arises from interactions between genetic predisposition and environmental exposures (exposome). This study used an integrative multi-omics to identify co-regulated genes in IPF fibroblasts. Differentially expressed miRNAs (DEMis) in lung fibroblasts from eight patients with IPF and four controls were defined by the threshold number of misclassifications <0.05, $p < 0.05$, fold change >1.2, and analyzed with DNA methylation (GSE107226) and mRNA expression datasets (GSE71351), respectively. miRNA–mRNA networks were constructed and analyzed for functional enrichment. Exposome-associated genes were identified through PubMed reviews. Five DEMis (*miR-143-3p*, *miR-138-5p*, *miR-152*, *miR-181a-5p*, *miR-181b-5p*) showed significant negative correlations with 43 differentially expressed genes (DEGs; $p < 0.05$). A network map visualized interactions between these 5 DEMis and their 43 target DEGs, highlighting strong miRNA–mRNA relationships. Integrative analysis of DNA methylation data identified 65 epigenetically regulated DEGs and 113 non-epigenetically regulated DEGs. Epigenetically regulated DEGs were enriched in cell motility, tissue development, and intercellular communication. Among 12 hub genes, *S100A4*, *PTGS2*, *FGF7*, *LEF1*, and *TFPI* were identified as potentially responsive to environmental risk factors such as cigarette smoke and particulate matter. This study demonstrates that coordinated regulation by miRNAs and DNA methylation modulates fibroblast gene expression in IPF, uncovering exposome-responsive pathways and potential therapeutic targets.

Keywords: integrated analysis, transcriptome, methylome, microRNAome, idiopathic pulmonary fibrosis

Introduction

Idiopathic pulmonary fibrosis (IPF) is a chronic and progressive interstitial lung disease marked by irreversible scarring of the lung parenchyma, which gradually impairs pulmonary function (Richeldi et al., 2017). Multiple factors are known to contribute to the pathogenesis of IPF, including genetic predisposition, environmental exposures (exposome), cellular and molecular alterations, and aging-related processes. Several studies have shown that environmental risk factors such as viral infections, micro-aspiration, tobacco smoke, air pollution, and occupational exposures are closely associated with the development and progression of IPF (Cui et al., 2023; Koo et al., 2017; Park et al., 2024; Sack and Raghu, 2019). To better understand how these diverse environmental factors influence disease development, the concept of the exposome has been introduced. The exposome represents the totality of nongenetic environmental exposures throughout the lifespan (Siroux et al., 2016; Vrijheid, 2014), serving as a complementary framework to the genome for understanding disease risk and progression. Importantly, environmental exposures can influence gene expression through epigenetic mechanisms, including DNA methylation, histone modifications, and microRNA (miRNA)-mediated regulation, thereby linking the exposome to IPF pathogenesis (Yang and Schwartz, 2015).

In our previous study, we analyzed DNA methylation and gene expression in IPF lung fibroblasts versus normal controls (GSE107226) (Lee et al., 2019). Approximately 5,850 CpGs (1.2%) mapped to 2,282 genes, with 34 differentially

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expressed genes (DEGs) showing significant correlations between DNA methylation and expression. Notably, Sanders et al. (2012) employed genome-wide DNA methylation arrays alongside mRNA microarrays in the same IPF lung tissue samples, identifying 16 genes whose differential DNA methylation inversely correlated with mRNA expression. However, only DNA methylation was considered, leaving other epigenetic factors unexplored.

Among various epigenetic regulators, miRNAs have emerged as key post-transcriptional modulators involved in the pathogenesis of IPF (Pandit et al., 2011). miRNAs are small noncoding RNAs composed of 18–25 bases that precisely regulate gene expression in the post-transcriptional phase through degradation or translation inhibition of the target mRNA. Several miRNAs function as dual regulators that may also affect gene transcription levels by indirectly regulating DNA methylation enzymes or transcription factors. To date, several miRNAs have been reported to be involved in various pathological pathways of IPF, such as TGF-β signaling, extracellular matrix reconstruction, and epithelial-mesenchymal metastasis, including *miR-21*, *miR-29*, *miR-200b/c*, *let-7d*, and *miR-199a-5p* (Zhou et al., 2024). Several studies have investigated miRNA–mRNA regulatory axes in IPF, demonstrating that specific miRNAs target key profibrotic or anti-fibrotic genes.

Although both miRNAs and DNA methylation are recognized contributors to IPF, their cooperative interactions remain insufficiently explored. *Nana-Sinkam* et al. (Dakhlallah et al., 2013) performed integrated profiling of miRNA expression and DNA methylation using matched human IPF lung tissue samples and fibroblasts, demonstrating an epigenetic feedback loop. Recently, there has been discussion about the interaction of IPF epigenetics, but integrated omics studies that simultaneously address miRNA and DNA methylation are still limited (Effendi and Nagano, 2023; Huang et al., 2025). While relatively few studies in IPF have jointly examined miRNA expression and DNA methylation in the same samples, cancer research has seen significant advances in this regard. For example, in hepatocellular carcinoma, a genome-wide analysis of miRNA expression and DNA methylation revealed inverse correlations for specific miRNAs such as *miR-18a* and *miR-125b-1* (Varghese et al., 2021), and a systems biology framework further highlighted DNA methylation–miRNA–mRNA regulatory networks relevant to metabolic and cell cycle pathways (Zahid et al., 2019). In lung adenocarcinoma, integrative analysis of DNA methylation and gene expression led to the development of a prognostic DNA methylation signature (Xu et al., 2022), while in glioblastoma, combined profiling identified hub genes with diagnostic potential via receiver operating characteristic (ROC) analysis (Alivand et al., 2021). These studies highlight how combining epigenetic and transcriptomic layers can reveal regulatory circuits and therapeutic targets that remain hidden in single-layer studies.

In the present study, we investigated the interplay among gene expression, miRNA levels, and DNA methylation in lung-derived fibroblasts from patients with IPF by applying such integrative approaches to provide novel insights into disease mechanisms and to uncover potential therapeutic targets. We further compare epigenetically and non-epigenetically regulated genes to identify distinct functional pathways, aiming to propose regulatory strategies for gene expression influenced by exposome-related factors.

Materials and Methods

Study subjects

Primary lung fibroblasts from subjects with IPF and controls were obtained from the Biobank of Soonchunhyang University Bucheon Hospital. The study protocol was approved by the Institutional Review Board (IRB) of Soonchunhyang University Seoul Hospital (IRB No. SCHUHIRB2025-07-004) and by the IRB of Soonchunhyang University (IRB No. 201910-BR-058). Written informed consent had been obtained by the biobank from all participants prior to sample collection.

RNA extraction and miRNA microarray analysis

Lung fibroblasts were isolated from lung tissue biopsies of eight patients with IPF and four control subjects with localized lung cancer, as previously described (Lee et al., 2017). Total RNA was extracted using TRIzol™ Reagent (Invitrogen, Carlsbad, CA) according to the manufacturer's instructions. For microarray analysis, 100 ng of total RNA was labeled with Cyanine-3 (Cy3) and hybridized to the SurePrint G3 Human miRNA Microarray (Release 19.0, 8 × 60K; Agilent Technologies, Palo Alto, CA), which profiles 2,006 human miRNAs. Hybridization and washing steps were conducted following the Agilent MicroRNA Microarrays Protocol Version 2.2. Microarray slides were scanned using the Agilent DNA Microarray Scanner, and signal intensities were extracted using Agilent Feature Extraction Software (v11.0.1.1).

Gene ontology analysis of functional pathways and protein–protein interaction network

To identify the biological functions and pathways of DEGs, we performed gene ontology (GO) enrichment analysis and Kyoto Encyclopedia of Genes and Genomes pathway analysis. DEGs were selected based on fold change ≥1.5 and p value <0.05. GO analysis was performed using WebGestalt (https://www.webgestalt.org/) (Zhang et al., 2005). Statistical significance was evaluated using false discovery rate (FDR) correction, and pathways with FDR <0.05 were considered significant. Protein–protein interaction (PPI) among DEGs was studied using STRING network analysis (https://string-db.org) with medium confidence interaction score of 0.4. Further, Cytoscape (3.9.1) was used to visualize the PPI network (León and Calligaris, 2017).

Statistical analysis

The gene expression data have been deposited in the NCBI Gene Expression Omnibus (GEO) under accession number GSE71351 (Lee et al., 2017). The CpG methylation data generated in this study are publicly available in the GEO database under accession number GSE107226 (Lee et al., 2019). For miRNA data, the R package DEGseq offers the option to use the Mann–Whitney test, which was used to compare beta values between the IPF and control groups. To assess the relationship between miRNA and gene expression,

Spearman's rank correlation analysis was conducted. A p value <0.05 was considered statistically significant throughout all analyses. The Threshold Number of Misclassifications (TNoM) score was calculated as previously described (Radom-Aizik et al., 2007). miRNAs were considered significantly differentially expressed if they exhibited an absolute fold change >1.2 and a p value <0.05, based on both TNoM scoring and Mann–Whitney test results between IPF and control fibroblasts. Differentially expressed miRNAs (DEMis) and DEGs were visualized using heat maps generated in GenomeStudio software (Illumina Inc., USA). The software is publicly available online at the Illumina website (https://www.illumina.com/products/by-type/informatics-products/microarray-software/genomestudio.html). To investigate miRNA–mRNA regulatory networks, the tool miRTarVis+ was used (L'Yi et al., 2017).

Results

Identification of differentially expressed miRNAs between IPF and controls

We profiled 2,006 miRNAs in pulmonary fibroblasts from patients with IPF ($n = 8$) and controls ($n = 4$), identifying 10 DEMis based on TNoM score <0.05, Mann–Whitney test $p < 0.05$, and fold change >1.2 (Fig. 1 and Table 1). Clinical characteristics of both groups were previously reported (Lee et al., 2017) (Table 2 and Supplementary Table S1). Target genes of DEMis were predicted using experimentally verified interactions and computational predictions via miRWalk 3.0 (Quillet et al., 2019). Among the 10 DEMis, 6 were predicted to target 64 out of 178 previously identified DEGs (Table 3) from our transcriptome analysis (GSE71351; Supplementary Table S2). Spearman's correlation analysis revealed that 5 DEMis (*miR-143-3p*, *miR-138-5p*, *miR-152*, *miR-181a-5p*, and *miR-181b-5p*) showed significant negative correlations ($p < 0.05$) with 43 DEGs (Table 4), indicating miRNA-mediated post-transcriptional repression.

Then, a network map was constructed to visualize relationships between these 5 DEMis and their 43 target DEGs (Fig. 2). Among the 43 DEMi-targeted DEGs, each miRNA showed distinct but overlapping target profiles. In particular, *miR-181a-5p* and *miR-181b-5p* shared 13 targets, including *TMTC1*, *MCC*, *ALDH3A2*, *RGS18*, *MYEF2*, *PF4V1*, *EPDR1*, *CABYR*, *PTGS2*, *FBN2*, *SIK2*, *FGF7*, and *CCL8*, suggesting cooperative regulation. Among the overlapping targets, *miR-181a-5p* and *miR-181b-5p* shared six genes with miR-143-3p, including *TMTC1*, *ALDH3A2*, *PTGS2*, and *FBN2*. In contrast, only one overlapping gene (*PTGS2*) was identified with *miR-152*, and a single gene (*FGF7*) overlapped with *miR-138-5p*. These findings suggest that overlapping regulation by multiple miRNAs may reinforce epigenetic control of key genes contributing to fibrotic remodeling in IPF.

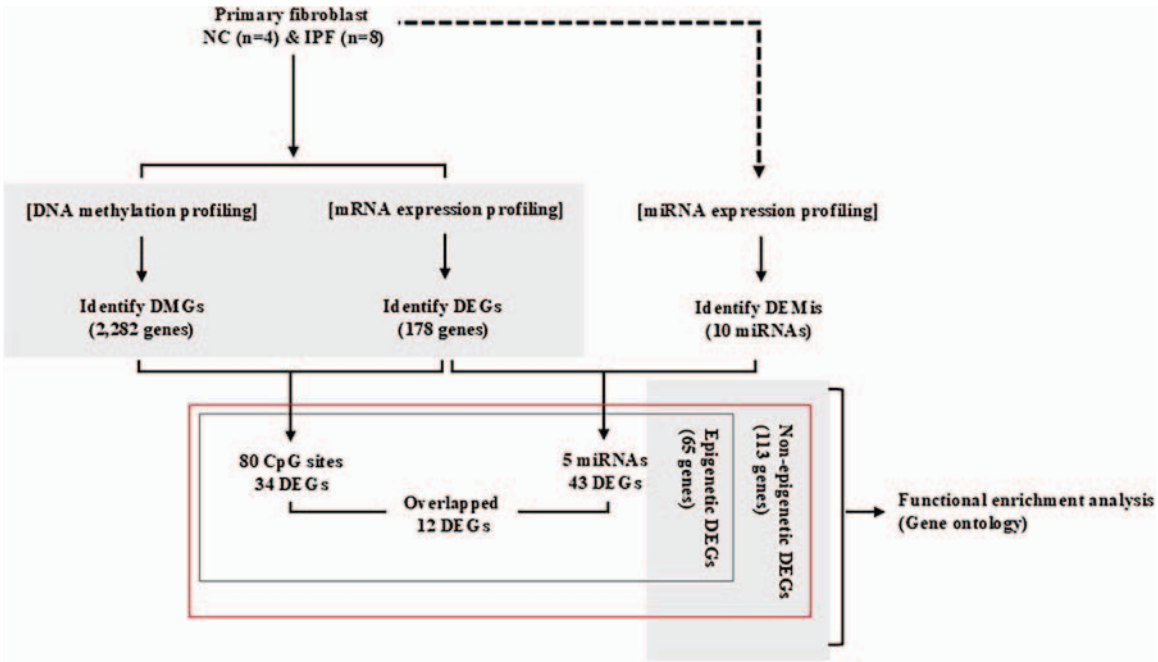

FIG. 1. Schematic workflow of the integrated multi-omics analysis. This pipeline demonstrates an integrative approach to identify epigenetically regulated genes between controls and IPF samples. miRNA expression profiling revealed 10 DEMis based on the following criteria: TNoM <0.05, t-test $p < 0.05$, and fold change >1.2. Of these, 6 DEMis were predicted to target 64 DEGs using target prediction databases. Among them, 5 DEMis showed significant negative correlations with 43 DEGs (Spearman correlation, $p < 0.05$). mRNA expression and DNA methylation data (GSE107226) were integrated to identify epigenetically co-regulated genes. This analysis revealed that 80 differentially methylated CpG sites were linked to 34 DEGs, suggesting transcriptional regulation via DNA methylation. Further integration with miRNA data identified DEGs potentially regulated by both methylation and miRNA. Common DEGs were subjected to GO analysis to investigate their biological roles and relevance as potential biomarkers or therapeutic targets in IPF. DEG, differentially expressed gene; DMG, differentially methylated gene; DEMi, differentially expressed miRNA; IPF, idiopathic pulmonary fibrosis; GO, gene ontology; TNoM, the threshold number of misclassifications.

Table 1. List of 10 microRNAs Differentially Expressed Between IPF and Controls

miRNA name	*Controls*	*IPF*	p *Value*	*TNoM*	*Fold change*
miR-138-5p	35.87 ± 2	7.64 ± 3.22	0.0002	0.004	–4.69
miR-6073	10.32 ± 2.49	2.46 ± 1.17	0.008	0.048	–4.2
miR-664a-3p	13.62 ± 0.38	7.94 ± 1.31	0.0031	0.048	–1.71
miR-143-3p	169.58 ± 12.44	106.12 ± 11.05	0.0056	0.048	–1.6
miR-4499	13.22 ± 0.18	9.1 ± 1.67	0.0435	0.004	–1.45
miR-181a-5p	161.92 ± 14.97	113.4 ± 7.5	0.0083	0.048	–1.43
miR-181b-5p	51.28 ± 4.32	36.7 ± 2.95	0.0181	0.048	–1.4
miR-4291	134.85 ± 15.82	98.11 ± 8.07	0.0422	0.048	–1.37
miR-152	75.43 ± 4.32	57.46 ± 3.55	0.0124	0.048	–1.31
miR-34a-5p	482.43 ± 39.78	592.31 ± 24.96	0.0346	0.048	1.23

Values are expressed as mean ± standard error of the mean (SEM) for normally distributed variables.
IPF, idiopathic pulmonary fibrosis; TNoM, the threshold number of misclassifications.

Functional divergence between epigenetically and non-epigenetically regulated DEGs

To explore regulatory interactions among miRNA expression, DNA methylation, and mRNA expression, we performed an integration analysis. We combined 43 DEGs predicted to be regulated by 5 DEMis with 34 DEGs predicted to be influenced by DNA methylation in our previous study (Lee et al., 2019). GO analysis was performed on two DEG groups: 65 DEGs with potential epigenetic regulation (miRNA targeting or DNA methylation) and 113 DEGs without such evidence (Supplementary Table S3). Epigenetically regulated 65 DEGs were enriched in biological processes related to cell motility and intercellular signaling, including positive regulation of cell migration, chemotaxis, tissue development, and positive regulation of cell communication, which are closely associated with fibrosis-related pathology such as fibroblast activation, tissue remodeling, and directed cell movement. In contrast, non-epigenetically regulated 113 DEGs were enriched in processes such as cytoskeleton organization, positive regulation of cellular component biogenesis, supramolecular fiber organization, plasma membrane-bounded cell projection assembly, regulation of extracellular matrix constituent secretion, cellular response to tumor necrosis factor, and regulation of pseudopodium assembly (Fig. 3 and Supplementary Table S4). Collectively, these findings suggest that epigenetic regulation exerts preferential influence on fibrosis-relevant cellular processes in IPF.

Table 2. Clinical Characteristics of the Study Subjects

Items	*Controls*	*IPF*
No.	4	8
Age (years)	54 (48–59)	64 (50–72)
Sex (male/female)	1/3	4/4
Smoke (CS/ES/NS)	0/0/4	0/4/4
Survival/death	ND	5/3
Follow-up duration (years)	ND	4.1 (2.1–7.0)
FVC (% pred.)	98 (87–106)	84 (56–99)[a]
DL_{CO} (% pred.)	90 (76–120)	60 (39–87)[a]

Values are expressed as median (interquartile range) for non-normally distributed variables. Differences between controls and IPF subjects were assessed using the Mann–Whitney U test.
[a]$p < 0.05$ compared with controls.
IPF, idiopathic pulmonary fibrosis; CS/ES/NS, current smoker/ex-smoker/never smoker; FVC, forced vital capacity; DL_{CO}, diffusing capacity of the lungs for carbon monoxide; ND, not determined.

Identification of exposome-associated epigenetic DEGs in IPF

To investigate epigenetically regulated DEGs potentially responsive to environmental exposures, the epigenetically regulated 65 DEGs were analyzed for GO enrichment terms associated with environmental responses, including positive regulation of cell migration, sprouting angiogenesis, locomotion, chemotaxis, tissue development, and intercellular communication (Fig. 3 and Supplementary Table S4).

Candidate genes involved in these processes were selected for further evaluation based on multiple lines of evidence: (1) differential expression in fibrotic lung tissues and experimental models, (2) response to environmental exposure in cellular or animal studies, and (3) gene–environment interactions or single nucleotide polymorphism (SNP) associations in GWAS/GWIS. Through this approach, several hub genes including *S100A4*, *PTGS2 (COX2)*, *FGF7*, *LEF1*, and *TFPI* were identified as potentially responsive to environmental risk factors such as smoking and particulate matter (PM2.5 or PM10) (Supplementary Table S5). Notably, these hub genes were among the 12 DEGs under dual epigenetic regulation (Fig. 1 and Supplementary Tables S6 and S7), suggesting that they may serve as key mediators linking environmental exposures to fibrosis-associated cellular pathways in IPF.

Comparison of functional enrichment between miRNA-targeted and methylation-associated genes in IPF

GO and pathway enrichment analyses of the 5 DEMis-targeted 43 DEGs revealed significant enrichment in biological processes related to bone mineralization, biomineral tissue development, and tissue morphogenesis. These genes were also implicated in heparin binding, respiratory system development, and alcohol metabolism. Furthermore, several GO terms were associated with the positive regulation of cell migration, motility, and locomotion, suggesting potential roles in cellular movement and differentiation during fibrotic remodeling (Supplementary Fig. S1 and Supplementary Table S8). Pathway analysis additionally demonstrated enrichment in the amplification and expansion of oncogenic pathways, hypothesized pathways in cardiovascular disease, and both

Table 3. List of Six microRNAs Differentially Expressed Between IPF and Controls

miRNA microarray				*Transcriptome*				*Spearman's rho*	
miRNA name	*miRNA fold change*	p *Value*	*TNoM*	*Gene name*	*Fold change*	p *Value*	*TNoM*	*Correlation coefficient*	p *Value*
miR-138-5p	−4.69	0.0002	0.004	*SLC25A45*	3.09	0.001	0.004	−0.76	0.004
		0.0002	0.004	*RNF144A*	2.47	0.017	0.048	−0.754	0.005
		0.0002	0.004	*FGF7*	11.21	0.007	0.004	−0.688	0.013
		0.0002	0.004	*RDH10*	3.81	0.01	0.048	−0.688	0.013
		0.0002	0.004	*ROR2*	22.13	0.034	0.048	−0.653	0.021
		0.0002	0.004	*CARD6*	2.04	0.01	0.048	−0.646	0.023
		0.0002	0.004	*RASL12*	8.07	0.039	0.048	−0.613	0.034
		0.0002	0.004	*LEF1*	3.61	0.009	0.048	−0.604	0.038
		0.0002	0.004	*TFPI*	2.22	0.003	0.048	−0.585	0.046
miR-143-3p	−1.6	0.0056	0.048	*PTGS2*	5.45	0.014	0.004	−0.944	4.E-06
		0.0056	0.048	*TSKU*	5.2	0.009	0.048	−0.93	1.E-05
		0.0056	0.048	*SIK2*	2.18	0.01	0.048	−0.909	4.E-05
		0.0056	0.048	*LOC286367*	6.45	0.024	0.048	−0.903	6.E-05
		0.0056	0.048	*FAM198B*	7.08	0.035	0.048	−0.895	8.E-05
		0.0056	0.048	*MOXD1*	4.06	0.008	0.048	−0.888	1.E-04
		0.0056	0.048	*ALDH3A2*	3.02	0.02	0.048	−0.888	1.E-04
		0.0056	0.048	*KCNJ2*	18.64	0.011	0.004	−0.881	2.E-04
		0.0056	0.048	*RGS18*	16.78	0.035	0.048	−0.862	3.E-04
		0.0056	0.048	*ARHGAP20*	4.01	0.014	0.048	−0.825	0.001
		0.0056	0.048	*TFPI*	2.22	0.003	0.048	−0.825	0.001
		0.0056	0.048	*FBN2*	8.67	0.006	0.048	−0.818	0.001
		0.0056	0.048	*TMTC1*	2.3	0.011	0.048	−0.797	0.002
		0.0056	0.048	*WDR52*	5.37	0.007	0.048	−0.766	0.004
		0.0056	0.048	*POSTN*	10.29	0.009	0.048	−0.761	0.004
		0.0056	0.048	*EPB41L3*	13.99	0.016	0.048	−0.619	0.032
miR-152	−1.31	0.0124	0.048	*PRKG2*	3.91	0.02	0.048	−0.79	0.002
		0.0124	0.048	*NUDT8*	2.29	0.016	0.048	−0.741	0.006
		0.0124	0.048	*DNER*	8.06	0.037	0.048	−0.727	0.007
		0.0124	0.048	*GAS1*	7.25	0.007	0.048	−0.706	0.01
		0.0124	0.048	*PTHLH*	2.67	0.027	0.048	−0.7	0.011
		0.0124	0.048	*EPB41L3*	13.99	0.016	0.048	−0.673	0.016
		0.0124	0.048	*NID1*	2.06	0.008	0.048	−0.671	0.017
		0.0124	0.048	*LOC100129104*	2.82	0.019	0.048	−0.664	0.018
		0.0124	0.048	*PTGS2*	5.45	0.014	0.004	−0.65	0.022
		0.0124	0.048	*C1QTNF9B*	3.43	0.005	0.004	−0.65	0.022
		0.0124	0.048	*RDH10*	3.81	0.01	0.048	−0.636	0.026
		0.0124	0.048	*ARHGAP20*	4.01	0.014	0.048	−0.608	0.036
		0.0124	0.048	*RNF144A*	2.47	0.017	0.048	−0.608	0.036
miR-181a-5p	−1.43	0.0083	0.048	*MCC*	5.98	0.037	0.048	−0.836	0.001
		0.0083	0.048	*RGS18*	16.78	0.035	0.048	−0.828	0.001
		0.0083	0.048	*ALDH3A2*	3.02	0.02	0.048	−0.811	0.001
		0.0083	0.048	*EPDR1*	2.89	0.001	0.004	−0.797	0.002
		0.0083	0.048	*KCTD12*	2.11	0.047	0.004	−0.783	0.003
		0.0083	0.048	*PTGS2*	5.45	0.014	0.004	−0.776	0.003
		0.0083	0.048	*FGF7*	11.21	0.007	0.004	−0.769	0.003
		0.0083	0.048	*CABYR*	2.51	0.026	0.048	−0.769	0.003
		0.0083	0.048	*RNF144A*	2.47	0.017	0.048	−0.769	0.003
		0.0083	0.048	*TMTC1*	2.3	0.011	0.048	−0.755	0.005
		0.0083	0.048	*D4S234E*	18.77	0.048	0.048	−0.753	0.005
		0.0083	0.048	*SIK2*	2.18	0.01	0.048	−0.692	0.013
		0.0083	0.048	*MME*	3.61	0.02	0.048	−0.657	0.02
		0.0083	0.048	*RDH10*	3.81	0.01	0.048	−0.657	0.02
		0.0083	0.048	*FBN2*	8.67	0.006	0.048	−0.643	0.024
		0.0083	0.048	*TFPI*	2.22	0.003	0.048	−0.629	0.028
		0.0083	0.048	*MYEF2*	4.81	0.01	0.048	−0.626	0.03
		0.0083	0.048	*CCL8*	22.84	0.038	0.004	−0.615	0.033
		0.0083	0.048	*PF4V1*	59.2	0.039	0.048	−0.615	0.033
		0.0083	0.048	*ING5*	2.28	0.01	0.048	−0.587	0.045
miR-181b-5p	−1.4	0.0181	0.048	TMTC1	2.3	0.011	0.048	−0.804	0.002
		0.0181	0.048	MCC	5.98	0.037	0.048	−0.798	0.002
		0.0181	0.048	ALDH3A2	3.02	0.02	0.048	−0.769	0.003
		0.0181	0.048	RGS18	16.78	0.035	0.048	−0.754	0.005
		0.0181	0.048	RBM47	2.03	0.027	0.048	−0.734	0.007
		0.0181	0.048	MYEF2	4.81	0.01	0.048	−0.734	0.007
		0.0181	0.048	*PF4V1*	59.2	0.039	0.048	−0.699	0.011
		0.0181	0.048	*EPDR1*	2.89	0.001	0.004	−0.685	0.014
		0.0181	0.048	*CABYR*	2.51	0.026	0.048	−0.685	0.014

(*continued*)

Table 3. (Continued)

miRNA microarray				Transcriptome				Spearman's rho	
miRNA name	*miRNA fold change*	p *Value*	*TNoM*	*Gene name*	*Fold change*	p *Value*	*TNoM*	*Correlation coefficient*	p *Value*
		0.0181	0.048	*PTGS2*	5.45	0.014	0.004	–0.671	0.017
		0.0181	0.048	*FBN2*	8.67	0.006	0.048	–0.636	0.026
		0.0181	0.048	*SIK2*	2.18	0.01	0.048	–0.608	0.036
		0.0181	0.048	*CCL8*	22.84	0.038	0.004	–0.601	0.039
		0.0181	0.048	*KCNJ2*	18.64	0.011	0.004	–0.594	0.042
		0.0181	0.048	*FGF7*	11.21	0.007	0.004	–0.594	0.042
miR-34a-5p	1.23	0.0346	0.048	*ITGA10*	–4.6	0.000	0.004	–0.559	0.059
		0.0346	0.048	*ZNF747*	–4.8	0.000	0.004	–0.552	0.063
		0.0346	0.048	*SCARA3*	–2.4	0.001	0.048	–0.545	0.067
		0.0346	0.048	*SUMO1*	–2.5	0.003	0.048	–0.392	0.208
		0.0346	0.048	*CADM1*	–3.3	0.003	0.048	–0.378	0.226
		0.0346	0.048	*CDH13*	–3	0.001	0.048	–0.357	0.255
		0.0346	0.048	*PTX3*	–3.7	0.043	0.048	–0.347	0.269
		0.0346	0.048	*ANKRD1*	–3.1	0.020	0.048	–0.267	0.401
		0.0346	0.048	*FAM30A*	–2.5	0.013	0.048	–0.259	0.417
		0.0346	0.048	*DTNA*	–2.1	0.002	0.048	–0.140	0.665

IPF, idiopathic pulmonary fibrosis; TNoM, the threshold number of misclassifications.

canonical and noncanonical Notch signaling. Other enriched pathways included fibrin clot (clotting cascade), Wnt signaling, regulation of lipolysis in adipocytes, and Ca^{2+} signaling. Notably, both canonical and β-catenin–independent Wnt signaling pathways were prominent among the enriched terms, indicating that Wnt-related signaling mechanisms may play central roles in regulating these genes (Supplementary Fig. S1 and Supplementary Table S8).

In contrast, GO and pathway enrichment analyses of the 34 DEGs associated with 80 differentially methylated genes highlighted biological processes related to embryonic development and morphogenesis, including labyrinthine layer and placental morphogenesis, embryonic limb and appendage morphogenesis, and cartilage development. Processes associated with chemotaxis, positive chemotaxis, taxis, sprouting angiogenesis, and enzyme-linked receptor protein signaling were also significantly enriched, reflecting developmental and angiogenic activities (Supplementary Fig. S2 and Supplementary Table S9).

Table 4. Five DEMis Between IPF and Controls and Their Corresponding 43 Target DEGs

miRNA	*Fold change*	*Gene list*
miR-143-3p	–1.6	*PTGS2*, *TSKU*, *SIK2*, *LOC286367*, *FAM198B*, *MOXD1*, *ALDH3A2*, *KCNJ2*, *RGS18*, *ARHGAP20*, *TFPI*, *FBN2*, *TMTC1*, *WDR52*, *POSTN*, *EPB41L3*
miR-152	–1.31	*PRKG2*, *NUDT8*, *DNER*, *GAS1*, *PTHLH*, *EPB41L3*, *NID1*, *LOC100129104*, *PTGS2*, *C1QTNF9B*, *RDH10*, *ARHGAP20*, *RNF144A*
miR-181a-5p	–1.43	*MCC*, *RGS18*, *ALDH3A2*, *EPDR1*, *KCTD12*, *PTGS2*, *FGF7*, *CABYR*, *RNF144A*, *TMTC1*, *D4S234E*, *SIK2*, *MME*, *RDH10*, *FBN2*, *TFPI*, *MYEF2*, *CCL8*, *PF4V1*, *ING5*
miR-181b-5p	–1.4	*TMTC1*, *MCC*, *ALDH3A2*, *RGS18*, *RBM47*, *MYEF2*, *PF4V1*, *EPDR1*, *CABYR*, *PTGS2*, *FBN2*, *SIK2*, *CCL8*, *KCNJ2*, *FGF7*
miR-138-5p	–4.69	*SLC25A45*, *RNF144A*, *FGF7*, *RDH10*, *ROR2*, *CARD6*, *RASL12*, *LEF1*, *TFPI*

IPF, idiopathic pulmonary fibrosis; DEG, differentially expressed gene.

Consistently, pathway analysis revealed enrichment in cell junction–related and Wnt-associated signaling pathways, such as adherens junction interactions, cell–cell junction organization, cell–cell communication, and Hippo signaling regulation pathways. Additionally, pathways involving splicing factor NOVA–regulated synaptic proteins and β-catenin–independent Wnt signaling (WP363, hsa04310) were enriched, suggesting that methylation-associated genes may contribute to intercellular communication and developmental signaling regulation in IPF fibroblasts (Supplementary Fig. S2 and Supplementary Table S9).

Together, these findings indicate that miRNA-targeted and methylation-associated genes display distinct yet complementary functional patterns, with the former mainly related to tissue morphogenesis and cell motility and the latter enriched in developmental and intercellular signaling pathways.

Discussion

In this study, we identified 10 DEMis in lung fibroblasts from patients with IPF compared with normal controls. Five of these miRNAs showed significant negative correlations with 43 DEGs, indicating post-transcriptional repression of fibrosis-related targets. Integration of miRNA and DNA methylation data further revealed 12 DEGs under dual epigenetic control, highlighting cooperative regulation of fibroblast activation and matrix remodeling. Raw microarray expression data for lung fibroblasts from IPF and control subjects are available in Supplementary Data.

Pathway analysis of these 12 DEGs (Supplementary Table S10) indicated associations with diverse signaling pathways, including the Extrinsic Pathway of Fibrin Clot Formation, β-catenin dependent and independent Wnt signaling, RUNX3-mediated Wnt signaling, TCF/LEF: CTNNB1 target gene binding, Ca^{2+} signaling, and endoderm differentiation (Supplementary Table S10). These results suggest that epigenetic regulation via both miRNA and DNA methylation may play a critical role in IPF-related pathological processes, including fibroblast activation, tissue remodeling, and cell migration, thereby extending previous

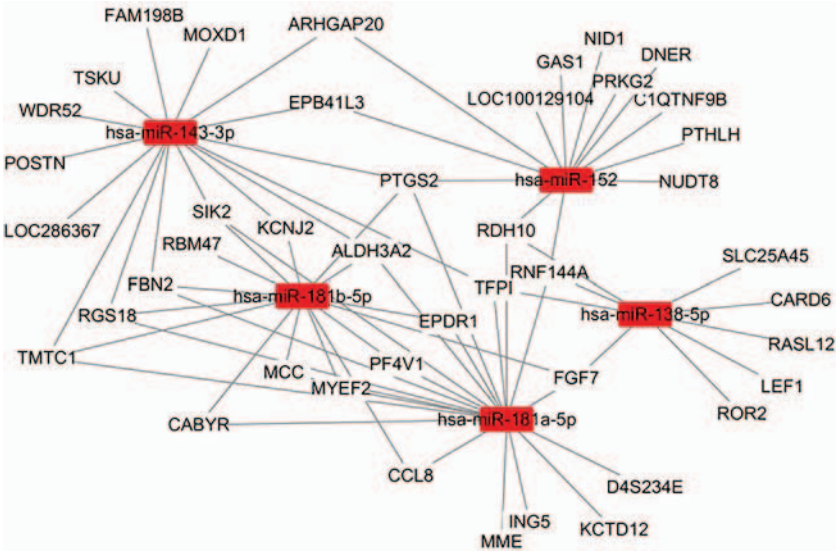

FIG. 2. The miRNA and mRNA regulatory network. The *red octagon* nodes represent the 5 DEMis, and the blue round rectangle nodes represent the 43 DEGs. The *gray lines* represent the networks predicted by miRWalk 3.0. DEG, differentially expressed gene; DEMi, differentially expressed miRNA.

reports (Lee et al., 2019; Mullenbrock et al., 2018) by providing direct fibroblast-level evidence for multilayered epigenetic control in IPF. Previous studies have demonstrated altered miRNA profiles and DNA methylation patterns in whole IPF lung tissue (Yang, 2012; Zhou et al., 2024), but few have explored their integration within the same cellular context. Our study uniquely applied an integrative multi-omics approach to primary fibroblasts, enabling the identification of coordinated regulatory circuits that would not be apparent from transcriptomic data alone.

Several identified miRNAs, including *miR-143-3p* and *miR-181* family members, have been implicated in fibroblast

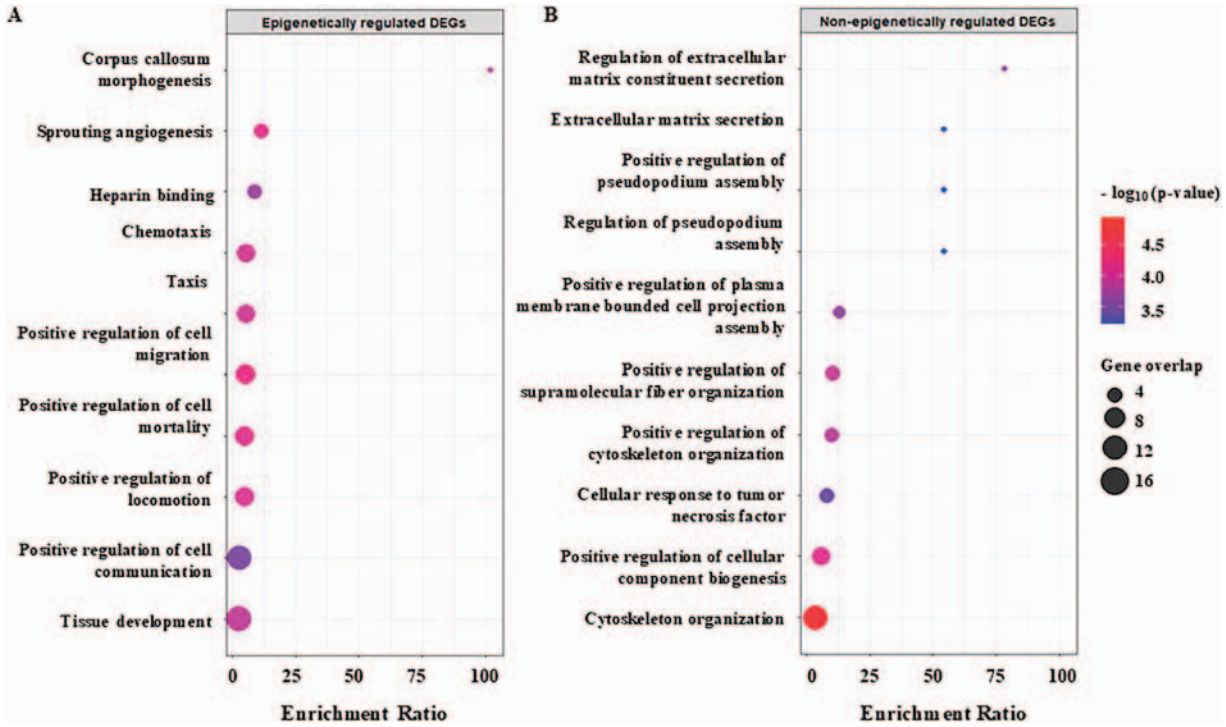

FIG. 3. Comparison of gene ontology enrichment between epigenetically regulated and unregulated DEGs. *Bubble plots* illustrating the top enriched biological processes in two groups of DEGs: **(A)** 65 DEGs regulated by either CpG methylation or miRNAs, and **(B)** 113 DEGs without epigenetic or post-transcriptional regulation. The *x*-axis represents the enrichment ratio, and the size of each bubble corresponds to the number of overlapping genes within the respective GO terms. The color gradient represents statistical significance ($-\log_{10} p$ value), ranging from *blue* (lower significance) to *red* (higher significance). DEG, differentially expressed gene; GO, gene ontology.

activation, epithelial–mesenchymal interactions, and extracellular matrix remodeling (Cadena-Suárez et al., 2022; Castaldi et al., 2020; Santos-Álvarez et al., 2022). Among these, miR-143-3p and miR-181b-5p were also reported to respond to environmental stimuli such as cigarette smoke, allergens, and particulate matter ($PM_{2.5}$/PM_{10}) (Pavanello et al., 2016; Pergoli et al., 2017; Rahman and Singh, 2020). This suggests that stress-responsive miRNAs may mediate the epigenetic effects of environmental exposures on fibroblast behavior, linking external stimuli to persistent fibrotic activation. The negative correlations observed between these miRNAs and their target DEGs, including *EPDR1*, *PTGS2*, and *FGF7*, suggest miRNA-driven repression of pro-fibrotic and inflammatory genes in IPF. Supporting this, previous studies have shown that EPDR1 is directly suppressed by *miR-429* (Zhao et al., 2021) and *miR-100-5p* (Liu et al., 2021b) in ovarian cancer, FGF7 is targeted by miR-155 in diabetic wounds, and PTGS2 is post-transcriptionally regulated by miRNAs, including miR-101-3p, in systemic sclerosis and rheumatoid arthritis (Moura et al., 2019; Wei et al., 2020; Xu and Chen, 2021). Collectively, these findings strongly support the central role of miRNAs in repressing key pro-fibrotic and inflammatory genes across diverse pathological contexts. Pathway analysis of the 5 DEMi-targeted 43 DEGs further supported this interpretation by highlighting canonical and noncanonical Notch, Wnt, and calcium signaling pathways, all of which have been previously implicated in fibroblast differentiation and tissue remodeling (Supplementary Fig. S1 and Supplementary Table S8). Together, these data underscore the importance of miRNA–mRNA regulatory networks as core molecular mechanisms driving fibroblast activation and fibrotic progression in IPF.

In parallel, comparative functional analysis revealed that miRNA-targeted genes and methylation-associated genes exhibit distinct yet complementary functional profiles. The miRNA-regulated genes were enriched in tissue morphogenesis- and motility-related processes, reflecting dynamic cellular responses during fibrotic remodeling, whereas methylation-associated genes were linked to developmental, angiogenic, and intercellular signaling pathways, suggesting reactivation of morphogenetic programs that may contribute to aberrant tissue regeneration in IPF (Supplementary Fig. S2 and Supplementary Table S9).

By integrating DNA methylation profiles with miRNA expression, we further distinguished between epigenetically regulated and non-epigenetically regulated DEGs. Epigenetically regulated 65 DEGs were preferentially enriched in biological processes associated with cell motility, chemotaxis, tissue development, and intercellular communication, processes intimately linked to fibroblast activation and matrix remodeling. In contrast, non-epigenetically regulated 113 DEGs were enriched in cytoskeletal organization, pseudopodium assembly, and structural biogenesis, suggesting that epigenetic regulation confers a selective advantage in controlling dynamic, motility-associated processes relevant to fibrosis. Notably, several epigenetically regulated DEGs such as *S100A4*, *FGF7*, and *LEF1* were positioned at the intersection of Wnt and calcium signaling pathways, reinforcing their mechanistic role in fibroblast activation. These results are consistent with previous studies demonstrating that multi-omics integration can reveal complex regulatory circuits not detectable in single-layer analyses (Alivand et al., 2021; Varghese et al., 2021; Xu et al., 2022; Zahid et al., 2019).

To further explore the connection between epigenetic changes and environmental responsiveness, we examined 65 DEGs subject to potential miRNA or methylation regulation (Supplementary Table S3). Several hub genes—*S100A4*, *PTGS2 (COX2)*, *FGF7*, *LEF1*, and *TFPI*—were identified as candidates potentially responsive to environmental stimuli such as cigarette smoke and particulate matter (Almutairi et al., 2025; Barua et al., 2002; Chen et al., 2023; Chen et al., 2010; Lee et al., 2016; Li et al., 2018; Li et al., 2023; Lim et al., 2019; Liu et al., 2021a; Martey et al., 2004; Reimann et al., 2015; Wang, 2023; Wei et al., 2025; Zhang et al., 2022) (Supplementary Table S5). These genes were selected based on evidence from fibrotic tissue expression patterns, experimental exposure models, and genome-wide association studies linking environmental risk factors to IPF susceptibility. Functional enrichment analysis revealed their involvement in cell migration, angiogenesis, chemotaxis, and inflammatory signaling, suggesting that environmental exposures may exacerbate fibrotic progression through epigenetic reprogramming of key fibroblast gene networks. By incorporating the exposome perspective, our study provides a mechanistic framework linking environmental risk factors to epigenetically regulated fibrosis pathways, thereby bridging environmental epidemiology and molecular pathology in IPF. The integrative nature of this work represents a conceptual advancement beyond single-omics approaches (Guo et al., 2014; Jeon et al., 2025). Our multilayer analysis reveals how DNA methylation and miRNA expression cooperatively shape fibrosis-relevant gene networks, and the exposome dimension further emphasizes external triggers that may modulate these epigenetic effects.

However, several limitations should be acknowledged. The sample size was limited, potentially affecting statistical power, and validation in independent cohorts and animal models is warranted. Moreover, as our analysis was confined to fibroblasts, future single-cell and spatial transcriptomic studies should evaluate whether similar epigenetic signatures exist across epithelial and immune populations. Direct exposure–response assays using patient-derived cells will also be necessary to establish causality. Lastly, while stringent statistical thresholds minimized false positives, they may have excluded miRNAs with subtle yet biologically meaningful effects. Future studies should thus integrate functional validation to confirm the relevance of candidate regulatory interactions.

Conclusion

Our multi-omics integration demonstrates that miRNA expression and DNA methylation cooperatively regulate gene networks central to fibroblast activation, tissue remodeling, and cell motility in IPF. Epigenetically regulated DEGs, including *S100A4*, *PTGS2*, and *FGF7*, appear to act as environmental response hubs linking exposure to fibrotic gene expression programs. These findings underscore the value of incorporating epigenetic and exposomic perspectives to unravel the complex molecular mechanisms driving

IPF and may inform the development of targeted therapeutic strategies aimed at reversing environmentally influenced epigenetic reprogramming.

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Authors' Contributions

Conceptualization: M.K.K., C.-S.P., and J.-U.L.; Data curation: M.K.K.; Investigation: M.K.K.; Validation: M.K.K.; Visualization: M.K.K.; Methodology: J.H.K. and H.G.H.; Resources: J.H.K., H.G.H., and H.S.C.; Formal analysis: H.S.C.; Software: J.H.K., H.G.H., and H.S.C.; Writing—original draft: M.K.K.; Writing—review and editing: J.H.K., H.G.H., C.-S.P., and J.-U.L.; Supervision: C.-S.P. and J.-U.L.; Project administration: H.S.C., C.-S.P., and J.-U.L.; Funding acquisition: M.K.K. and J.-U.L. All authors have read and approved the final version of the article.

Disclosure Statement

No competing financial interests exist.

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Supplementary Material

Supplementary Data
Supplementary Tables
Supplementary Figure S1
Supplementary Figure S2

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